

**Section A**

Answer **all** questions.

Refer to the separate Resource Folder to answer the following questions.

1. (a) Use the information under **Figure 1** to answer the following question.  
The wave speed of ultrasound in body tissue is 1540 m/s.

Use the equation:

$$\text{wavelength} = \frac{\text{wave speed}}{\text{frequency}}$$

to calculate the wavelength of the ultrasound used in a 2 MHz scan.

[3]

wavelength = ..... m

- (b) Use the information in **Figure 4** to answer the following questions.

- (i) Select the most suitable radioisotope of **iodine** from the table in **Figure 4** that would be used as a tracer. Explain your choice.

[2]

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- (ii) The photograph shows a patient undergoing external beam radiation therapy for a brain tumour.



Select the most suitable radioisotope from the table in **Figure 4** that would be used for this treatment. Explain your choice.

[3]

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- (c) Refer to **Figures 4 and 6** in the Resource Folder to answer the following question.

State which of the radioisotopes **R1, R2, R3, R4, R5** or **R6**, is lutetium-177.  
**Explain** your reasoning.

[3]

Radioisotope ..... is lutetium-177.

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